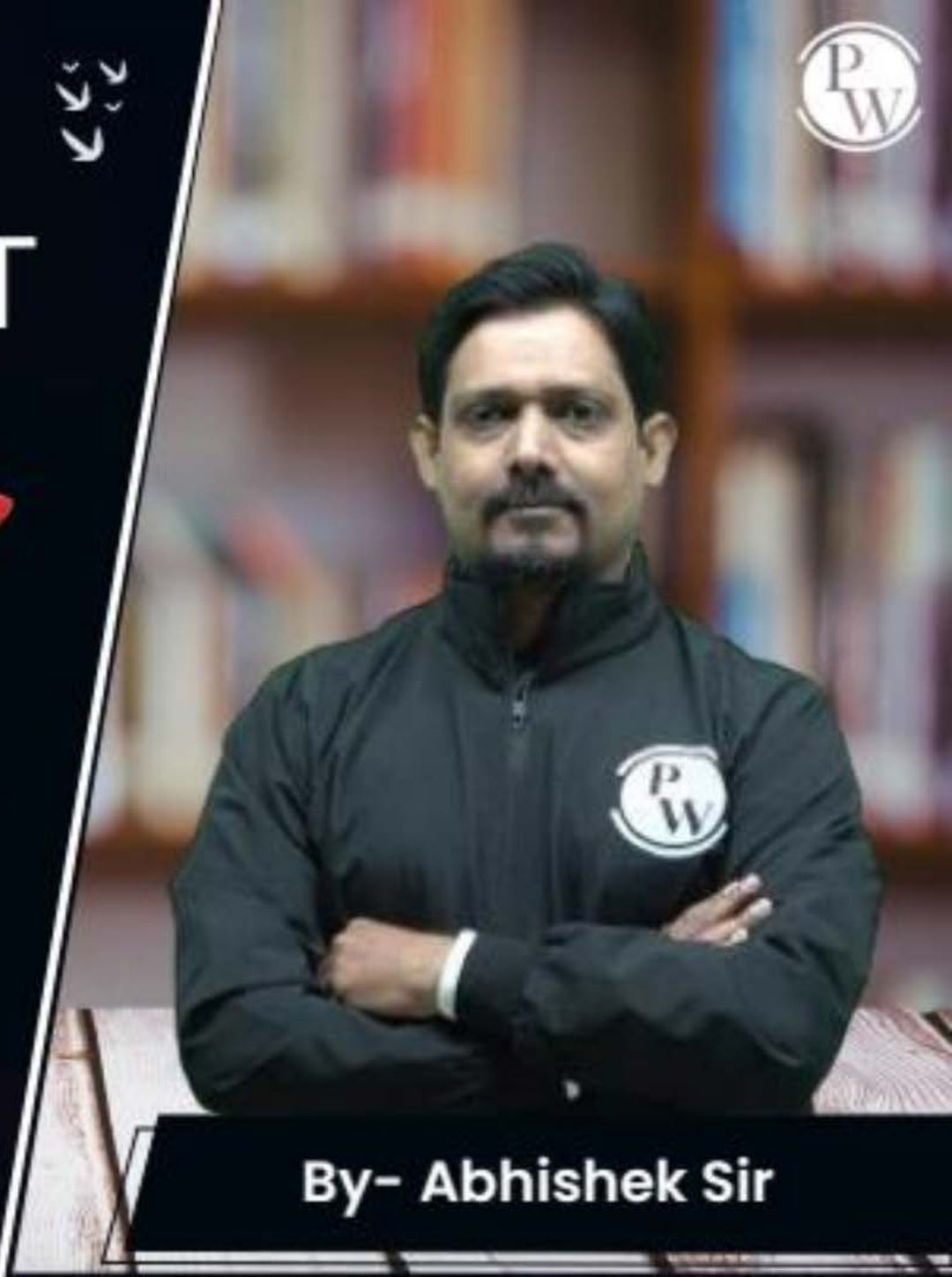
Computer Science & IT



C programming

Data Types & Operator

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



7pm-9pm

Topic	Structure of c program
Topic	type of statement
Topic	Statement break Token
Topic	Identifier, keywords, constant, operator, special
Topic	Character string

Slide

Topics to be Covered



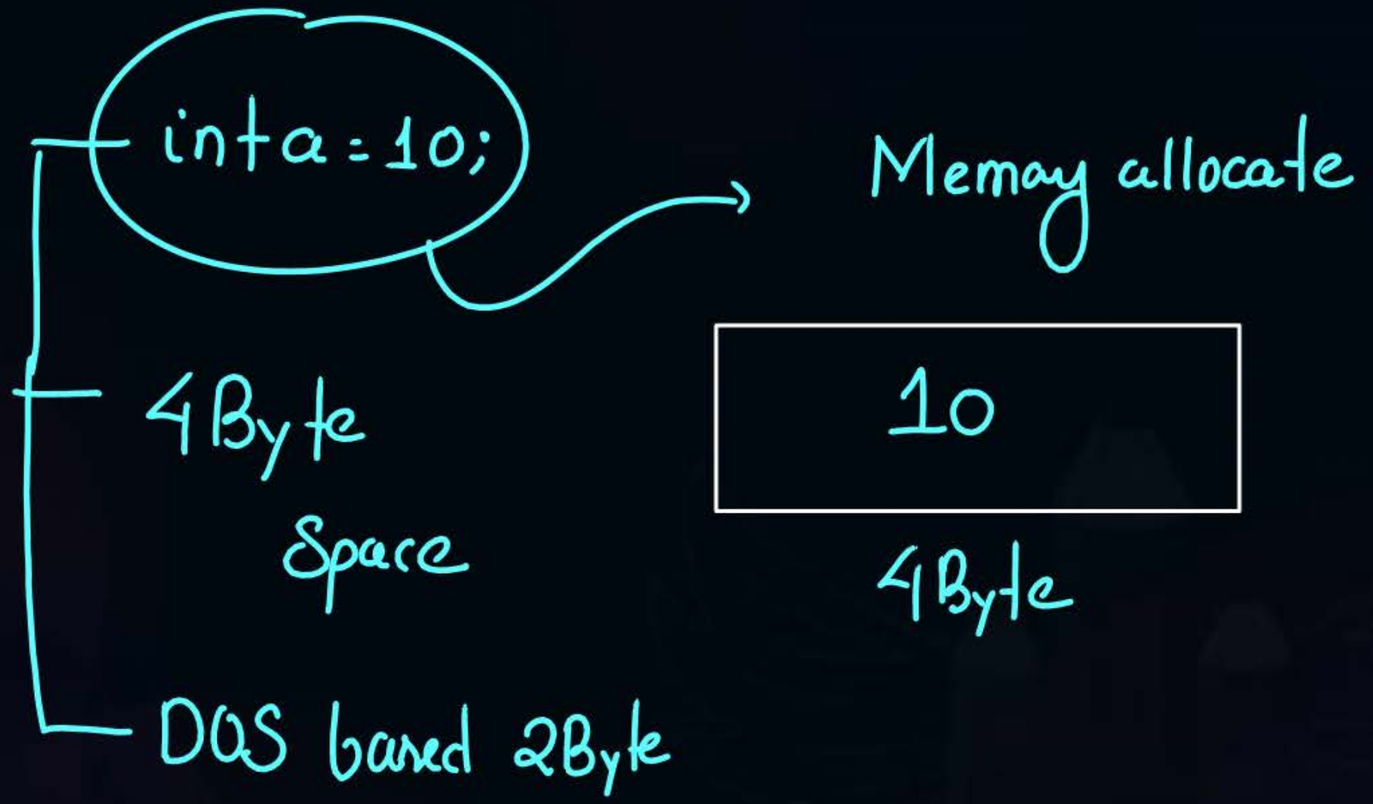
- Topic — Basic Data type
- Topic — %d, %u (Singed & unsigned)
- Topic — operators
- Topic — Arthmetic expression
- Topic

Slide



Topic : Primitive Data Types

- int
- character
- float



Slide



Topic : Integer Data Type

Every Data type is having a size

int :- 4Byte = 4 × 1 Byte = 4 × 8 = 32 bits

character : 1 Byte 1 × 8 bits = 8 bits

float : 4Byte 4 × 8 bits 32 bits

Double : 8 Byte 8 × 8 bits 64 bits

Slide



Topic : Integer Data Type

modifiers

int

- unsigned . Height
- Signed . profit / loss

Slide



Topic : Integer Data Type

Second modifier

int
- short :- 2 Byte
- long : 8 Byte

if int is 2B (long 4B, long long 8B)



Topic : Integer Data Type

Unsigned No. 1001 binary No.

$1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 9$

4 bits unsigned

15 — | 1 | 1 | 1 | 1 | - unsigned

Slide



Topic : Unsigned Integer Representation

```
int a=10;
printf("%u", a);
```

4 bit unsigned value

Minimum 0000 - 0

Max 1111 - 15



← 4 bits

Binay to decimal

10

Slide



Topic : Unsigned Integer Representation

$$1111$$

$$1\times2^3+1\times2^2+1\times2^1+1\times2^0$$

$$8+4+2+1=15$$

Slide



Topic : Unsigned Integer Representation

8 bits

Min 00000000 — 0

11111111
76543210

$2^7 \times 1 + 2^6 \times 1 + 2^5 \times 1 \times 2^4 \times 1 + 2^3 \times 1$

$+ 2^2 \times 1 + 2^1 \times 1 + 2^0 \times 1$

$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$

Max - 11111111 — 255

Slide



Topic : Unsigned Integer Representation

n bits ✓

Min — 000 · · · 0

Max 111 · · · 1

$= 2^n - 1$

formula – $\frac{a(r^n-1)}{r-1}$

1 1 1 1 · · · 1 1 0

$(n-1)\ (n-2)$ ... 1 0

$2^{n-1} + 2^{n-2} + 2^{n-3} + \cdots + 2^1 + 2^0$ (GP series)

$= \frac{2^0(2^n-1)}{2-1} = 2^n - 1$

Slide



Topic : Unsigned Integer Representation

Range of unsigned No- $0 \text{ —— } 2^n - 1$

unsinged int Size : 4B = 32 bit

$0 \text{ —— } 2^{32} - 1$

Short unsigned int Size : 2B : $0 - 2^{16} - 1$

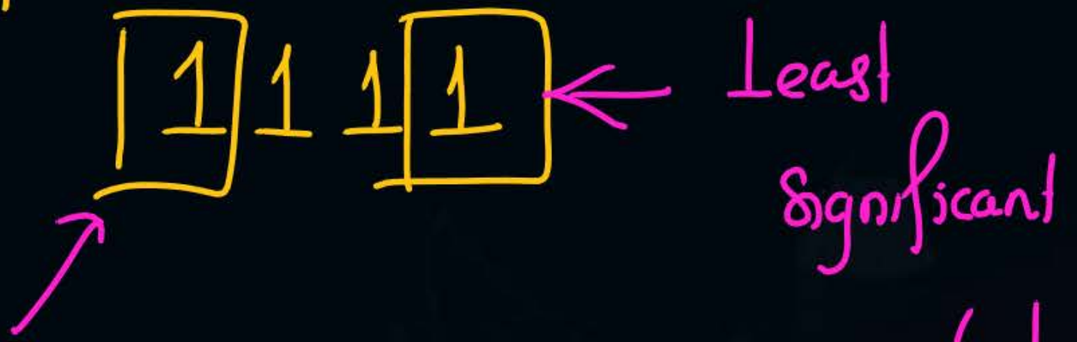
Slide



Topic : Signed Interger Representation

Singed No. int a = -10;

Singed No. 2's complement



Slide



Topic : Range of 2's complement representation

MSB is reserved for sign

MSB — 0 then +ve No.

MSB - 1 then -ve No

Slide



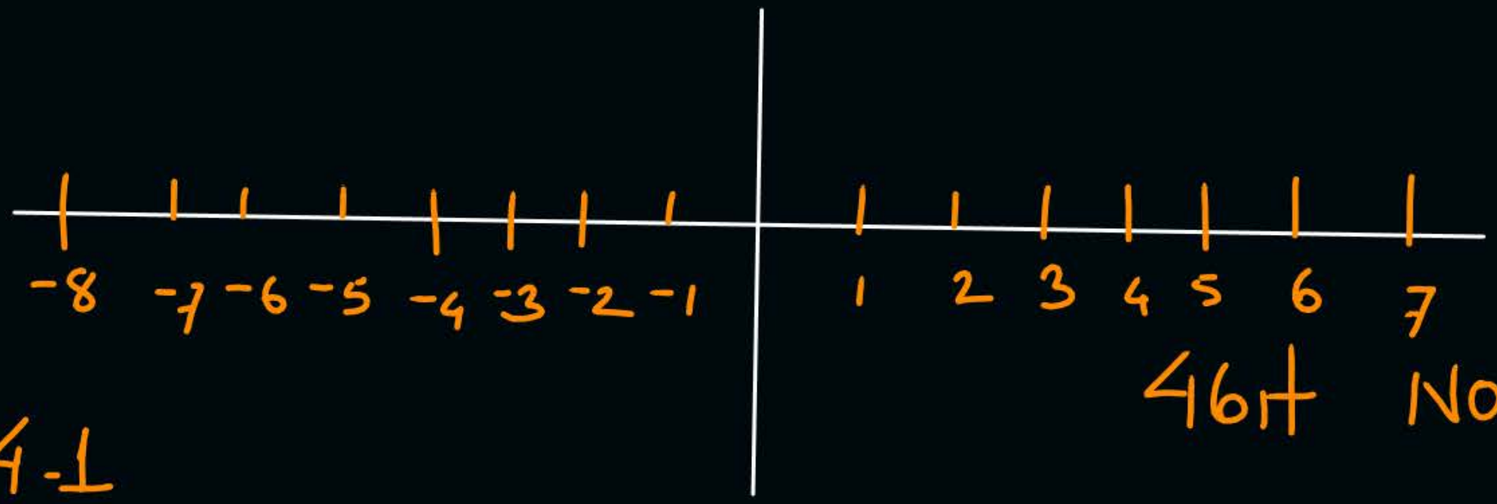
Topic : Range of 2's complement representation

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111

+ve

1000	$-1\times2^3+0+0+0 = -8$
1001	$-1\times2^3+0+0+1\times2^0 = -7$
1010	$-2^3+2^1 = -6$
1011	-5
1100	$= -2^3+2^2 = -8+4 = -4$
1101	$= -2^3+2^2+0+2^0 = -8+4+1 = -3$
1110	-2
1111	$-1\times2^3+1\times2^2+1\times2^1+1\times2^0 = -8+4+2+1 = -1$

Slide



4 bit No

-2^{4-1}

positive range

$0 - 2^{4-1} - 1$

4 bits

Range :- -2^{4-1} to $2^{4-1} - 1$

n bits

$$-2^{n-1} \text{ to } 2^{n-1} - 1$$

Singed int —— 4Byte = 32 bits

-2^{31} to $+2^{31}-1$

Short int —— 2Byte = 16 bits

-2^{15} to $2^{15}-1$



Topic : Range of 2's complement representation

1001 - Singned No

3 2 1 0

$-1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

$= -8 + 0 + 0 + 1 = -7$

Singed No



Topic : Range of 2's complement representation

1011 Singed No.

$-1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

$-8 + 0 + 2 + 1 = -5$

Slide



Topic : Range of 2's complement representation

$$\underset{4\,3\,2\,1\,0}{1\,0\,1\,1\,0} \quad \text{Singed No}$$

$$-1\times2^4+0\times2^3+1\times2^2+1\times2^1+0\times2^0$$

$$-16+0+4+2+0 : -16+6 : -10$$

Slide



Topic :

What is the range of 8 bit unsigned Integer

A. 0 to 256

B. 1 to 256

C. 0 to 255

D. 1 to 255



Topic : Question

What is the range of 7 bit unsigned Integer

A. 0 to 128

B. 0 to 127

C. 1 to 128

D. 1 to 127



Topic : Question

What is the range of 8 bit signed Integer

A. 0 to 255

B. -128 to 127

C. -127 to 127

D. -126 to 128

Slide



Topic : Question

What is the range of 16 bit signed Integer

A. 0 to 65536

B. -32768 to 32767

C. -32767 to 32767

D. -16384 to 16384

-2^{15} to $2^{15}-1$

-32768 to 32767

2^9 - 512

2^{10} - 1024

2^{11} - 2048

2^{12} - 4096

2^{13} - 8192

2^{14} - 16384

2^{15} - 32768

2^{16} - 65536

Slide



Topic : Question

What is the range of n bit signed Integer

A. -2^n to 2^{n-1}

B. -2^{n-1} to $2^{n-1}-1$

C. $-2^n -1$ to 2^{n-1}

D. $-2^n +1$ to 2^{n-1}



Topic : Question

What is the range of n bit unsigned Integer

A. 0 to 2^{n-1}

B. 0 to $2^{n} - 1$

C. 0 to $2^{n-1} + 1$

D. $-2^{n} + 1$ to 2^{n-1}

Slide



Topic : Question

What is the decimal value of unsigned 6 bit number

100000

A. 64

B. 32

C. 63

D. 31

Slide



Topic : Question

What is the decimal value of signed 6 bit number

100000

A. -64

B. -32

C. 64

D. 32

Slide



Topic : Question

What is the decimal value of unsigned 6 bit number

100011

A. 64

B. 32

C. 35

D. 31

100011
543210

$1\times2^5+0+0+0+2^1+2^0$

$=32+2+1=35$

Slide



Topic : Question

What is the decimal value of signed 6 bit number 100011

A. 64

B. 29

C. -29

D. 31

Slide



Topic : Question

What is the decimal value of unsigned 8 bit number

1000 0011

A. 131

B. 128

C. 259

D. 255

Slide



Topic : Question

What is the decimal value of signed 8 bit number 1000 0011

A. -125

B. -131

C. 259

D. -259

1000 0011
7654 3210

$-1 \times 2^7 + 1 \times 2^1 + 1 \times 2^0$

$-128 + 2 + 1 = -128 + 3$

$= -125$

Slide



Character

t me/Abhisheksharoma.pw

Arithmetic operators

7 - 9

youtube

Data structure

Classes

9pm

Slide



2 mins Summary

Topic integer

Topic short long

Topic Singned unsigned

Topic Range

Topic

Slide

THANK - YOU

Slide